

Solid Waste Management Data Analysis

Pune Ward-Level Waste Generation Analysis — Project Report

1. Executive Summary

This project analyzes a real Pune solid-waste dataset containing ward-level household counts and reported daily waste quantities. The main objective is to compare total waste with a household-normalized metric, Waste per Household (kg/day). This provides two complementary views of waste burden.

Key takeaway: Total waste and waste per household identify different priority areas. Some wards have high overall waste because of their scale, while a smaller number show unusually high waste per household and require further investigation.

2. Problem Statement & Objectives

Municipal solid-waste planning requires understanding both where total waste volumes are highest and where waste generation is unusually high relative to household counts.

- Evaluate waste generation across Pune wards.
- Compare total waste quantity across wards.
- Calculate Waste per Household (kg/day).
- Identify potential statistical outliers using IQR.
- Study relationships between household count and waste metrics.
- Classify wards using total waste and household-normalized waste.
- Develop recommendations supported by the available data.

Scope correction: Vehicle transit time, segregation leakage, collection timing and real-time reporting were excluded because these operational measurements are not present in the Pune dataset used here.

3. Dataset Overview & Data Cleaning

The dataset contains 42 ward-level records. Core variables used were Ward name, Number of HH, and Waste quantity (Tonnes Per Day). The original data was preserved and a separate analysis copy was used.

Variable	Description	Type
Ward name	Ward/area name	Categorical
Number of HH	Number of households	Integer
Waste quantity (Tonnes Per Day)	Reported daily waste quantity	Numeric

The dataset also contained a segregation-households column. It was identical to Number of HH for all 42 rows, so it was treated as redundant and excluded from analysis rather than being used to claim 100% segregation.

Derived metric: Waste per Household (kg/day) = Waste quantity (Tonnes/Day) × 1000 ÷ Number of HH.

4. Exploratory Data Analysis

4.1 Summary Statistics

Metric	Minimum	Mean	Median	Maximum
Total waste (TPD)	14.64	37.915	38.34	66.53
Waste/HH (kg/day)	0.1245	0.4717	0.4021	1.6681

Waste per Household is concentrated toward lower values with a longer right tail. An exact numerical skewness coefficient was not calculated; the distribution comment is based on the observed values and histogram.

4.2 Correlation Analysis

Variables	Pearson r	Interpretation
Households vs Total Waste	0.043	Very weak linear relationship
Households vs Waste/HH	-0.693	Moderately strong negative linear relationship

The household-count versus total-waste correlation is approximately 0.043, indicating a very weak linear relationship in this sample. This does not prove that household count has no effect on waste generation.

The -0.693 relationship between household count and Waste/HH must be interpreted cautiously because household count is the denominator of the derived metric; part of the relationship can therefore be mathematical rather than causal.

4.3 Outlier Detection Using IQR

For Waste per Household, Q1 = 0.2663 kg/day, Q3 = 0.5363 kg/day, and IQR = 0.2700 kg/day. The upper outlier boundary is approximately 0.9412 kg/day.

Ward	HH	Total Waste (TPD)	Waste/HH (kg/day)
Janta Vasahat-Duttawadi	15,753	23.57	1.4962
Warje-Malwadi	35,058	39.14	1.1164
Kondhwa Khurd-Mittha Nagar	27,378	45.67	1.6681

These are potential statistical outliers, not confirmed data errors. They should be validated using additional field or operational data.

4.4 Ward Categorization Matrix

Using the median total waste of 38.34 TPD and median Waste/HH of 0.4021 kg/day:

Category	Wards
High Total Waste + High Waste/HH	18
High Total Waste + Low Waste/HH	3
Low Total Waste + High Waste/HH	3
Low Total Waste + Low Waste/HH	18

5. Key Findings & Insights

- **Highest total waste:** Kondhwa-Yelwalewadi, Dhankawadi-Sahakanagar, Hadapsar-area combined ward record reported 66.53 TPD.
- **Highest Waste/HH:** Kondhwa Khurd-Mittha Nagar reported 1.6681 kg/household/day.
- **Other potential outliers:** Janta Vasahat-Duttawadi (1.4962) and Warje-Malwadi (1.1164).
- **Different priorities:** high total waste and high Waste/HH do not always occur in the same wards.
- **Household count is not a strong linear predictor of total waste:** $r \approx 0.043$.
- **Data-quality issue:** the segregation-households field duplicated household count and was not used as a segregation-performance measure.

6. Actionable Recommendations

- **Capacity allocation:** use total TPD to guide collection capacity, manpower, vehicles and treatment/disposal planning.
- **Targeted investigation:** validate the three high Waste/HH outlier wards with field-level information.
- **Two-metric planning:** use total waste for capacity planning and Waste/HH for identifying unusual local patterns.

- **Collect operational data:** vehicle trips, vehicle capacity, collection frequency, route distance/time, missed collections and fuel/operating data.
- **Expand waste composition data:** wet/dry waste, recyclable waste, commercial waste and independently measured segregation rates.
- **Future dashboard:** build a Power BI or Python dashboard for ward-level totals, normalized waste and operational indicators.

7. Limitations

- The analysis uses aggregated ward-level data rather than household-level records.
- The dataset does not provide sufficient time-series information for detailed seasonal analysis.
- The segregation field is redundant in these records.
- Correlation does not establish causation.
- Potential outliers require validation with additional source data.
- Vehicle performance, route efficiency and collection-time claims are outside this dataset's scope.

8. Technical Stack

Component	Technology / Method
Programming	Python
Data manipulation	Pandas
Visualization	Matplotlib
Environment	Jupyter Notebook
Methods	Descriptive statistics, correlation, IQR, ranking, median classification

9. Project Workflow

1. Load the real Pune dataset.
2. Inspect rows, columns, data types and data-quality issues.
3. Preserve original data and create an analysis copy.
4. Exclude the redundant segregation field.
5. Calculate Waste per Household.
6. Perform descriptive statistics and rankings.
7. Create visualizations.
8. Calculate Pearson correlations.
9. Detect potential outliers using IQR.
10. Classify wards using median-based quadrants.
11. Interpret findings and develop evidence-based recommendations.

10. Conclusion

This project demonstrates how Python and Pandas can transform a real solid-waste dataset into useful analytical information. Total waste and household-normalized waste provide different but complementary views of ward-level waste burden. Three wards were identified as potential high outliers in Waste/HH, while the four-quadrant classification provides a simple framework for distinguishing capacity requirements from areas needing targeted investigation.

The project also demonstrates an important data-analysis principle: recommendations should be based on what the data actually measures. Adding richer operational, composition, demographic and time-series data would enable a more complete solid-waste decision-support system.